DISPLAY BACKLIGHT & AMOLED POWER IC LINE MAP

Display Driver & High-Voltage Circuitry | Sheet 1 of 2

AMOLED DUAL POLARITY VOLTAGE RAILS

Display Bias Generator (TI TPS65633 / SGM3833)

AVDD (Analog VDD): +4.6V DC @ 150mA | Filter Cap C2401 | Diode: 480mV

ELVDD (OLED Positive Rail): +4.6V DC @ 400mA | Output Coil L2401 | Diode: 490mV

ELVSS (OLED Negative Rail): -1.4V to -4.0V DC | Inverting Charge Pump Cap C2405

SW_BOOST (Switching Node): 2.2MHz PWM drive | Diode: 550mV (cathode to rail)

MIPI DSI Data Pairs: D0P/D0N, D1P/D1N, D2P/D2N, D3P/D3N (All: 380mV +/- 15mV)

If ELVSS rail reads 0V or positive voltage, replace OLED bias IC immediately

CONFIDENTIAL - SALEN MOBILE GURU CHIP-LEVEL HARDWARE LAB

DISPLAY BACKLIGHT & AMOLED POWER IC LINE MAP

Display Driver & High-Voltage Circuitry | Sheet 2 of 2

LCD BACKLIGHT DRIVER CIRCUIT

Step-Up LED Driver Boost (up to 40V DC)

LED_A (Anode String): +18V to +28V DC under load (3.8V standby) | Diode: 430mV

LED_K1 (Cathode String 1): Current sink PWM controlled (0.3V - 0.8V)

LED_K2 (Cathode String 2): Current sink PWM controlled (0.3V - 0.8V)

Schottky Diode D2401: Forward drop 180mV - 240mV | Reverse breakdown > 45V

Boost Inductor L2402: 10uH 1.8A Shielded SMD Power Inductor (DCR < 0.15 Ohm)

Common Fault: Blown backlight filter resistor (0 Ohm) causing black display

CONFIDENTIAL - SALEN MOBILE GURU CHIP-LEVEL HARDWARE LAB